

Diagnostic performance of the measurement of nitric oxide in exhaled air in the diagnosis of COPD phenotypes



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ABSTRACT

Introduction: COPD is a highly heterogeneous disease that has a serious impact on affected populations. Patients share some of the features of bronchial asthma, often summarized under the term ACOS (asthma-COPD overlap syndrome). The objective of this study is to evaluate the utility of the measurement of nitric oxide in exhaled air (F_ENO₅) in the diagnosis of COPD phenotypes.

Material and methods: The study comprised cross-sectional observation of patients receiving assistance during a respiratory outpatient visit. Patient data was collected on lung function, F_ENO, questionnaires scoring CAT® (COPD Assessment Test), and COPD clinical phenotype.

Results: 192 patients were studied: 103 with COPD; 16 healthy non-smokers; 30 healthy smokers; and 43 asthmatics. COPD patients were grouped by phenotype: 34 non exacerbators (33.0%); 22 ACOS (21.3%); 13 frequent exacerbators with emphysema (12.6%); and 34 frequent exacerbators with chronic bronchitis (33.0%). ACOS patients showed significantly higher F_ENO₅ values compared to the others after adjustments for confounding factors. F_ENO₅ demonstrated greater diagnostic accuracy than the bronchodilator test (BT) in the diagnosis of ACOS and COPD phenotypes (AUC 0.79 vs 0.74), with an optimal cut-off value of 19 ppb (sensitivity 0.68, specificity 0.75).

Conclusions: Patients with COPD have different levels of F_ENO₅ depending on the COPD phenotype. F_ENO₅ measurement provides better diagnostic accuracy than BT, with an optimal cut-off value of 19 ppb.

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1. Introduction

COPD is a widely occurring disease with significant associated morbidity and mortality. About 10% of the adult population in Spain were affected in 2010 [1], while COPD is the third leading cause of death worldwide [2]. At present it is considered a heterogeneous disease for which patients cannot be only categorized on the basis of functional parameters [3]. In this context, the concept of COPD

phenotype has been generalized in recent years. This takes into consideration additional information and facilitates more personalized treatments in accordance with the clinical presentation of patients' symptoms [4].

In the assessment of COPD, recognition of the disease's heterogeneity has led to the development of guidelines and strategies which include factors not exclusively related to lung function, such as the clinical impact of the disease [5]. Further, the Spanish guidelines for COPD (GesEPOC) [6] proposes treatment for four phenotypes: non-exacerbators with emphysema or chronic bronchitis; frequent exacerbators with chronic bronchitis; frequent exacerbators with emphysema; and asthma-COPD overlap syndrome (ACOS) [7].

Although bronchial asthma and COPD have different pathogenic mechanisms, namely inflammatory, genetic and immunological airway patterns [8–11], there is a subgroup of patients with COPD who share mechanisms of both conditions, for which this differentiation is complex. They are now known as ACOS patients

Abbreviation list: COPD, Chronic Obstructive Pulmonary Disease; ACOS, Asthma-COPD overlap syndrome; F_ENO₅, fraction of nitric oxide in exhaled air measured at constant rate of 50 mL/s; Ppb, parts per billion; CAT, COPD assessment test questionnaire; ROC, receiving operator curves; BMI, body mass index; ICS, inhaled corticosteroids; FEV1postBD, forced expiratory volume in 1s postbronchodilator; FVCpostBD, forced vital capacity post bronchodilator; Δ FEV1, improvement in forced expiratory volume in one second (FEV1) after 400 µg salbutamol.

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[12–15]. To identify patients with ACOS, some approaches have been published in the literature, but none of these have been validated or replicated in different scenarios and this fact has led to controversies in how these patients should be diagnosed [16], although the majority of these approaches are based on the detection of eosinophilic bronchial inflammation. For example, the Spanish Guidelines for COPD (GesEPOC) suggests a number of factors that may indicate eosinophilic airway inflammation and an increased likelihood of response to inhaled corticosteroid treatments [12] and, to date, is the only one that has been validated in real practice [17]. However, some of these factors included in the proposal, such as sputum eosinophilia, are difficult to measure in a conventional assessment, whereas others that are easier to measure, such as the determination of fraction of nitric oxide in exhaled air ($F_{E}NO_{50}$) are not included. Several papers have shown that $F_{E}NO_{50}$ is associated with the presence of eosinophils in sputum, bronchial hyperresponsiveness, and a better response to inhaled corticosteroid treatments [18,19] suggesting perhaps that $F_{E}NO_{50}$ is a marker for ACOS.

The objective of this study is to evaluate the diagnostic performance of $F_{E}NO_{50}$ measurement in exhaled air for phenotype diagnosis in a cohort of COPD patients who received outpatient assistance.

2. Material and methods

A cross-sectional observational study was conducted on patients who received assistance consecutively in an outpatient pulmonary care facility. The individuals involved met the following eligibility criteria: they were 40 years of age or older and were suitable for a lung function test; showed no respiratory disease other than asthma and COPD; had not experienced episodes of respiratory infection in the 6 weeks prior to the visit; and had not used any oral corticosteroids in the previous 4 weeks. Patients were divided into four study groups: patients with COPD; healthy non-smokers; healthy smokers; and patients affected by bronchial asthma.

COPD patients met the following inclusion criteria: they should have proven cumulative consumption of at least 10 pack/year, with a diagnosis of COPD based on the recommendations in the GOLD strategy [5] (bronchodilator FEV1/FVC ratio <0.70) guidelines; were in a stable phase of their disease with no exacerbation in the previous 4 weeks; and had not participated in any other research study or clinical trial. Patients were classified according to GOLD 2011 strategy in ABCD groups and clinical phenotype as proposed by GesEPOC: non-exacerbators with emphysema or chronic bronchitis; frequent exacerbators with chronic bronchitis; frequent exacerbators with emphysema; and ACOS. ACOS was defined according to the validated criteria if one of this major criteria was present (previous history of asthma/wheezing outside chest infections or a documented very positive bronchodilator test (>14% and >400 mL)) or if two of this minor criteria were present (blood eosinophil count >3%, IgE levels >100 UI/L, 2 documented positive bronchodilator test with >12% or 200 mL gain in FEV1, atopy or previous history sensitization to aeroallergens demonstrated by positive skin prick test or specific IgE to allergens).

The study was approved by the Reference Committee of ethics and clinical research (CEI- Granada) and was developed in accordance with the ethical standards laid down in the Declaration of Helsinki of 1975. All participants gave written informed consent.

For each study participant, data was collected on demographics, usual drug treatment, and smoking status, and a functional respiratory exploration (spirometry, lung volume testing and diffusion test) was performed with a bronchodilator test (400 µg of salbutamol) according to published protocols [20] (Jaeger MasterScreen Body-Diff®, CareFusion, Germany).

According to international recommendations [21], $F_{E}NO_{50}$ measurement took place prior to spirometry, between 8 and 11 am. Two reproducible determinations of $F_{E}NO_{50}$ were collected for all patients at a constant flow of 50 mL/s with a chemiluminescence analyzer of nitric oxide (HypAirFeNO®, Medisoft, Belgium). Patients were invited to avoid the consumption of coffee or tobacco in the previous 12 h, as well as inhaled medications.

For each patient with COPD, clinical notes recorded the number and severity of exacerbations in the previous year (moderate ones defined as the need for oral corticosteroids and/or antibiotics and the severe ones defined as hospitalization for COPD), the cumulative consumption of tobacco, and the CAT® [22] self-administered questionnaire in Spanish. The person responsible for the assessment of each of the participants was blind to the result of the determination of $F_{E}NO_{50}$.

2.1. Statistical analysis

Means and standard deviations of quantitative variables, or median and interquartile range when appropriate, as well as sample size, are reported. For non-normal variables, the absolute and relative frequencies of qualitative variables are reported. Continuous variables were compared using the Student's *t*-test for those normally distributed (using Kolmogorov–Smirnov test to determine goodness of fit to the normal distribution) and the Mann–Whitney U or Kruskal–Wallis tests for those who not. The Chi-squared test was used to compare categorical variables, and the associated *p*-value was given.

To compare $F_{E}NO_{50}$ levels among COPD phenotypes, a non-parametric Kruskal–Wallis test was performed.

The accuracy of the bronchodilator test and $F_{E}NO_{50}$ measurement, respectively, was compared for the diagnosis of COPD phenotype. ROC curves (receiving operator characteristics) were built and AUC (area under curve) was calculated for each method. Sensitivity and specificity tests were used to establish the optimal cut-off values for $F_{E}NO_{50}$ measurement.

Tables, lists, analyses and graphics were produced using SPSS v18. A value of *p* < 0.05 was considered statistically significant.

3. Results

Of 219 candidates who participated in the study, 192 were eligible. They were divided into healthy non-smokers (8%), healthy smokers (15. %), patients with bronchial asthma (22%), and patients with COPD (53%) (Fig. 1). The characteristics of the study population are summarized in Table 1. There was a little proportion of patients without respiratory disease (healthy smokers and non smokers), but ICS use was detected also in those groups. Patients with bronchial asthma showed higher $F_{E}NO_{50}$ levels compared to the other groups, and patients with COPD showed higher $F_{E}NO_{50}$ levels compared to non-smoking healthy controls (Fig. 2).

The study population included 103 patients with COPD. Most were male (88.3%) with an average age of 71 (±7.9) years and an average post-bronchodilation FEV1 of 57% (±17) predicted.

Table 2 shows the characteristics of patients with COPD according to GOLD 2011 groups. There were differences across the groups in terms of pulmonary function, frequency of moderate and severe exacerbations, use of ICS and CAT scores. No differences were observed in $F_{E}NO_{50}$ levels between the GOLD 2011 groups, as expressed in Fig. 3.

When GesEPOC Guidelines were applied, patients were divided into ACOS (21.3%), frequent exacerbators with chronic bronchitis (33.0%), frequent exacerbators with emphysema (12.6%) and non-exacerbators with chronic bronchitis or emphysema (33.0%). Table 3 shows the characteristics of patients with COPD according

to GesEPOC guidelines. Patients with ACOS had a greater bronchodilator response and were more frequently treated with ICS, and also have more frequently history of asthma and atopy. There were no differences among phenotypes in CAT scores, or in the degree of bronchial obstruction.

ACOS patients had significantly higher $F_E\text{NO}_{50}$ values than those with other COPD phenotypes, even after adjustment for active smoking and the use of ICS, as shown in Fig. 4. If the active smokers were excluded from the analysis, the results showed also differences between COPD phenotypes [median values 23 ppb (ACOS) vs 16 (non exacerbator with emphysema) vs 15 (for frequent exacerbator with chronic bronchitis) vs 16 (for frequent exacerbators with emphysema), $p = 0.006$].

By analyzing the relationship between the bronchodilator test and $F_E\text{NO}_{50}$ measurement in COPD patients, a significant correlation was observed between the two measurements, mild ($r = 0.22$, $p = 0.026$) as shown in Fig. 4.

$F_E\text{NO}_{50}$ measurement in the diagnosis of ACOS exhibited superior diagnostic accuracy compared to the bronchodilator test with an area under the higher curve (0.793 vs. 0.746), as shown in Fig. 5. Using the various selection options, the optimal cut-off value for the diagnosis of ACOS using $F_E\text{NO}_{50}$ was 19 ppb, with a sensitivity of 0.68 and a specificity of 0.75 (Fig. 6).

4. Discussion

The results of this study show that the measurement of $F_E\text{NO}_{50}$ may be more useful than a bronchodilator test in the diagnosis of COPD phenotype in patients who receive assistance at a respiratory clinic, with an optimal cut-off value of 19 ppb. Although this levels are in the normal range for the general population [23], data on FENO distribution in COPD patients are scarce and could not be extrapolated from general population.

These results are similar to those published by other groups, which showed a relationship between $F_E\text{NO}_{50}$ and bronchodilator response [24,25] and between $F_E\text{NO}_{50}$ and the presence of bronchial hyperresponsiveness, a fact that is considered the

distinguishing feature of bronchial asthma [16]. The optimal cut-off value for the diagnosis of ACOS is similar to that described by other researchers. Some [26] observed an average level of 20 ppb $F_E\text{NO}_{50}$ in COPD patients with a previous diagnosis of bronchial asthma. Others [27] found a relationship between $F_E\text{NO}_{50}$ of 19 ppb and sputum eosinophils ($\geq 3\%$) in a series of patients hospitalized for COPD exacerbation and [28] in predicting eosinophilic airway inflammation in patients with chronic cough. Our study shares results with these studies, but differs from them in terms of the characterization of ACOS with strict criteria (that are not all related to the previous history of asthma, as another guidelines do) published before and included in GesEPOC guidelines, allowing the clinician to differentiate COPD phenotypes with simple and non invasive tests than can be used in ambulatory assistance.

We founded correlation between $F_E\text{NO}_{50}$ and bronchodilator response, but this correlation doesn't imply that the clinician should choose from one of those, as they can offer complementary information.

The percentage of patients with ACOS differs slightly from that reported elsewhere [29,30], where the prevalence was lower, with figures close to 15%. This may be explained by the different study populations recruited in various health care settings, or by the prevalence of bronchial asthma in the study countries.

These results may be helpful in the diagnosis of ACOS when it comes to replacing a test that is not habitually available in most centers, such as sputum eosinophilia. Although in various studies the latter was associated with $F_E\text{NO}_{50}$ and its response to ICS [19], these results have not been validated in this population of ACOS. Furthermore, although blood eosinophilia could be a good marker of ICS response in terms of exacerbation prevention [31], to date no data exists comparing blood eosinophils and $F_E\text{NO}_{50}$ in COPD.

This study shows some weaknesses, including the presence of active smokers in the study population (28% of the total sample), although there were no differences between phenotypes in smoking habits. In addition, the use of ICS can decrease the levels of $F_E\text{NO}_{50}$. This suggests that the cut-off values of $F_E\text{NO}_{50}$ for the diagnosis of ACOS could not be extrapolated to a "naïve" treatment

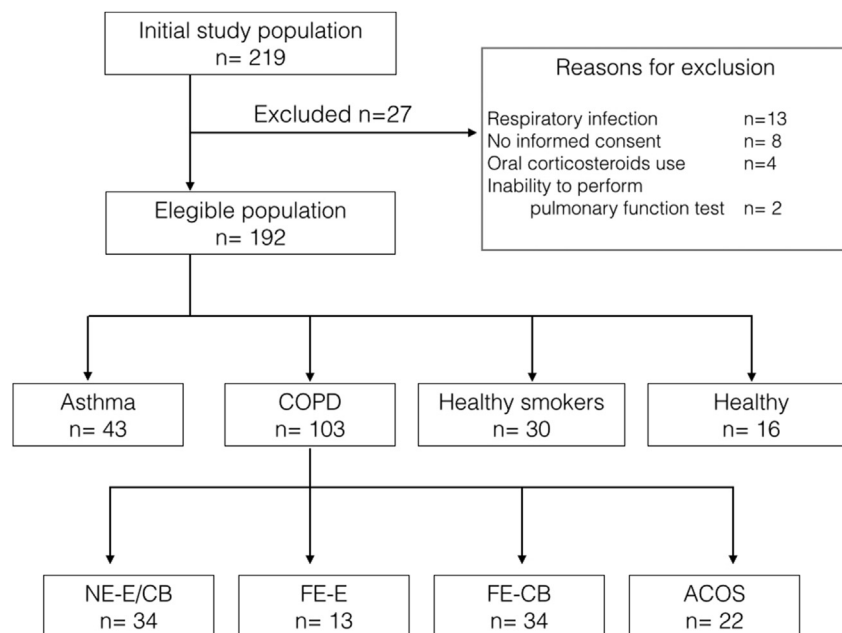


Fig. 1. Flowchart of Study Population. NE-E/CB: non-frequent exacerbators with chronic bronchitis or emphysema. FE-E: frequent exacerbators with emphysema. FE-CB: frequent exacerbators with chronic bronchitis. ACOS: asthma-COPD overlap syndrome.

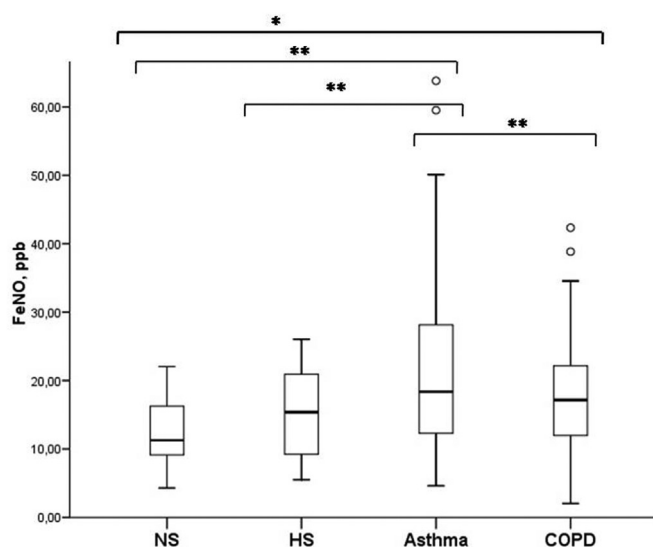
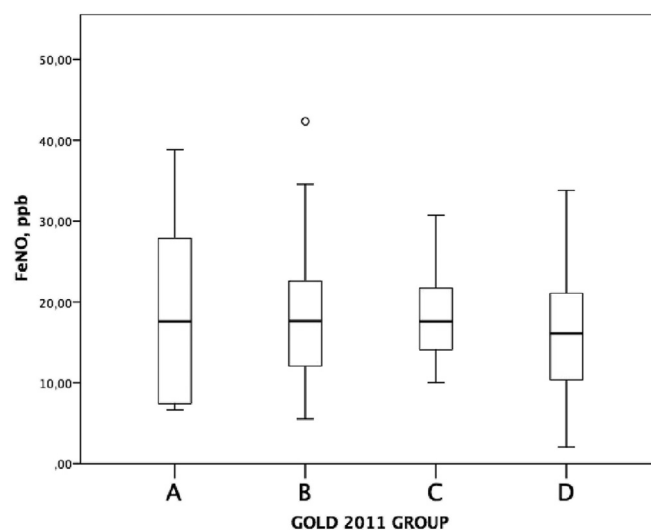
Table 1
Characteristics of study population.

	Healthy non- smokers (n = 16)	Healthy smokers (n = 30)	Asthma n(=43)	COPD (n = 103)
Age, yrs	57.9 ± 17.0	67.8 ± 9.8	54.5 ± 16.1	70.9 ± 12.5
Sex, M/F	6/10	24/6	10/33	91/12
BMI	30.6 ± 6.4	29.2 ± 6.5	31.7 ± 6.3	30.0 ± 4.7
Current smoker, n (%)	0 (0%)	30 (100%)	2 (4.6%)	29 (28.1%)
ICS use, n(%)	3 (18.8%)	14 (46.7%)	35 (81.4%)	62 (60.1%)
Pulmonary function test				
FEV ₁ postBD, (% pred)	95 ± 19	96 ± 3	90 ± 16	60 ± 21
FVC postBD, (% pred)	95 ± 17	88 ± 18	93 ± 16	83 ± 19
FEV ₁ /FVC	81 ± 7.6	80 ± 1.3	80 ± 7.2	54 ± 13
ΔFEV ₁ , (%)	5.9 ± 4.2	8.9 ± 9.9	6.2 ± 5.8	8.5 ± 8.2
Reversibility, n (%) ^a	2 (12.5%)	9 (30.0%)	9 (20.9%)	27 (26.2%)
F _E NO ₅₀ , ppb (median, IQ range)	12 [7,17]	15 [9,20]	22 (8, 36)	17 [25,9]

BMI: body mass index; COPD: chronic obstructive pulmonary disease; ICS: inhaled corticosteroids; FEV₁postBD: forced expiratory volume in 1s postbronchodilator; FVCpostBD: forced vital capacity post bronchodilator; ΔFEV₁: improvement in forced expiratory volume in one second (FEV₁) after 400 µg salbutamol, F_ENO₅₀: level of exhaled nitric oxide measured at 50 mL/s.

Values are expressed as mean (±standard deviation) unless mentioned otherwise.

^a Defined as improvement in FEV₁ >200 mL and >12% after 400 µg salbutamol.

**Fig. 2.** Levels of Exhaled Nitric Oxide measured at constant rate of 50 mL/s (F_ENO₅₀) across Study Groups. *: p < 0.05 and **: p < 0.01 (Kruskal–Wallis test). NS: healthy non-smoker. HS: healthy smoker. COPD: chronic obstructive pulmonary disease.**Fig. 3.** Level of Exhaled Nitric Oxide measured at constant rate of 50 mL/s (F_ENO₅₀) among COPD groups of GOLD 2011 strategy.**Table 2**
Characteristics of Patients with COPD (according to GOLD 2011 strategy groups).

	GOLD A (n = 6)	GOLD B (n = 32)	GOLD C (n = 10)	GOLD D (n = 54)	p value
Age, yrs	65 ± 9	72 ± 9	67 ± 13	71 ± 9	0.330
Sex, M/F	5/1	27/5	10/0	48/6	0.582
BMI	34 ± 4.5	30 ± 6.2	29 ± 4.3	29 ± 4.6	0.123
Smoking history					
Current smoker, n (%)	2 (33.3%)	10 (31.2%)	2 (20.0%)	15 (27.8%)	0.906
Pack- yrs	45 ± 8.8	41 ± 16.7	44 ± 19.9	47 ± 15.6	0.463
ICS use, n(%)	1 (16.7%)	13 (40.6%)	7 (70.0%)	39 (72.2%)	0.004
Pulmonary function test					
FEV ₁ postBD, (% pred)	71 ± 7.1	68 ± 14.3	52 ± 11.2	49 ± 16.9	<0.001
FVC postBD, (% pred)	95 ± 9.4	88 ± 17.0	77 ± 12.6	74 ± 18.3	0.001
Reversibility, n (%) ^a	2 (33.3%)	11 (34.4%)	1 (10.0%)	13 (24.4%)	0.436
ΔFEV ₁ , (%)	15.4 ± 12.1	7.4 ± 7.1	5.0 ± 4.6	7.2 ± 6.9	0.378
Exacerbations, moderate & severe, previous yr	0.50 ± 0.54	0.50 ± 0.50	2.50 ± 0.84	2.57 ± 1.28	<0.001
CAT® score	8.0 ± 2.6	20.1 ± 5.4	7.2 ± 1.5	22.1 ± 6.0	<0.001
F _E NO ₅₀ , ppb (median, IQ range)	18 [12,28]	18 [24,14]	18 [23,15]	16 [22,9]	0.668

BMI: body mass index; COPD: chronic obstructive pulmonary disease; ICS: inhaled corticosteroids; FEV₁postBD: forced expiratory volume in 1s postbronchodilator; FVCpostBD: forced vital capacity post bronchodilator; ΔFEV₁: improvement in forced expiratory volume in one second (FEV₁) after 400 µg salbutamol, F_ENO₅₀: level of exhaled nitric oxide measured at 50 mL/s.

Values are expressed as mean (±standard deviation) unless mentioned otherwise.

^a Defined as improvement in FEV₁ >200 mL and >12% after 400 µg salbutamol.

Table 3

Characteristics of Patients with COPD (according to clinical phenotypes as defined by GesEPOC guidelines).

	ACOS (n = 22)	FE-CB (n = 34)	FE-E (n = 13)	NE-CB/E (n = 34)	p value
Age, yrs	68 ± 8.0	72 ± 10.4	69 ± 11.2	71 ± 9.9	0.440
Sex, M/F	20/2	32/2	10/3	29/5	0.354
BMI	30 ± 4.72	30 ± 3.61	25 ± 4.25	31 ± 6.60	0.004
Asthma history, n (%)	5 (22.7%)	2 (2.9%)	1 (7.7%)	2 (5.8%)	0.002
Atopy, n (%)	6 (27.2%)	3 (8.8%)	0 (0.0%)	4 (11.7%)	0.043
Blood eosinophils, %	3.8 ± 2.5	2.4 ± 1.0	3.0 ± 1.6	2.9 ± 1.5	0.053
Smoking history					
Current smoker, n (%)	5 (22.7%)	9 (26.5%)	2 (15.4%)	13 (38.2%)	0.327
Pack- yrs	47 ± 17	45 ± 17	48 ± 13	42 ± 15	0.560
ICS use, n (%)	16 (72.7%)	28 (76.5%)	8 (61.5%)	10 (28.6%)	0.001
Pulmonary function test					
FEV ₁ postBD, (% pred)	58 ± 17	50 ± 16	49 ± 12	65 ± 17	0.003
FVC postBD, (% pred)	83 ± 16	75 ± 17	77 ± 21	85 ± 18	0.079
FEV ₁ /FVC	55 ± 10	51 ± 11	50 ± 7	54 ± 15	0.186
F _E NO50, ppb (median, IQ range)	23 [16,32]	14 [8,20]	16 [8,24]	15 (8(22)	<0.001
Reversibility, n (%)*	14 (63.6%)	6 (17.6%)	1 (7.7%)	6 (17.6%)	<0.001
ΔFEV ₁ , (%)	15 ± 12.1	7.4 ± 7.1	5.0 ± 4.6	7.2 ± 6.9	0.001
ΔFEV ₁ , (L)	0.19 ± 0.16	0.08 ± 0.08	0.05 ± 0.05	0.09 ± 0.09	<0.001
Exacerbations, moderate & severe, previous yr	1.7 ± 1.4	2.7 ± 1.0	2.5 ± 1.3	0.53 ± 0.56	<0.001
CAT® score	18 ± 7.1	21 ± 8.2	19 ± 5.5	17 ± 7.2	0.228

NE-CB/E: non-frequent exacerbators with chronic bronchitis or emphysema. FE-E: frequent exacerbators with emphysema. FE-CB: frequent exacerbators with chronic bronchitis. ACOS: asthma-COPD overlap syndrome.

BMI: body mass index; COPD: chronic obstructive pulmonary disease; ICS: inhaled corticosteroids; FEV₁postBD: forced expiratory volume in 1s postbronchodilator; FVCpostBD: forced vital capacity post bronchodilator; ΔFEV₁: improvement in forced expiratory volume in one second (FEV₁) after 400 µg salbutamol, F_ENO50: level of exhaled nitric oxide measured at 50 mL/s.

Values are expressed as mean (±standard deviation) unless mentioned otherwise.

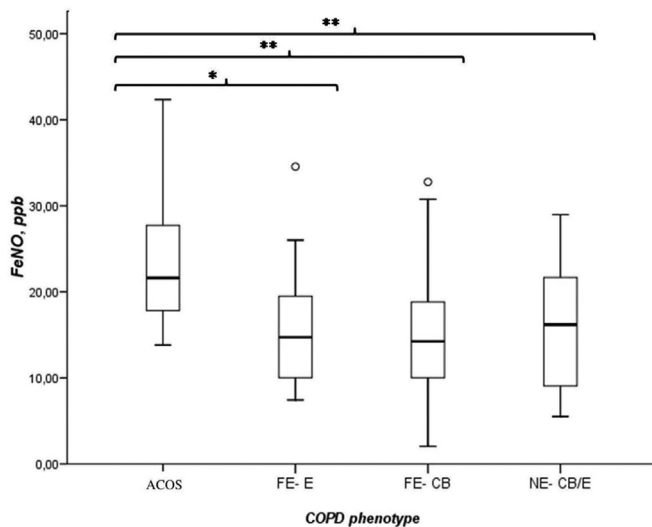


Fig. 4. Level of Exhaled Nitric Oxide (measured at constant rate of 50 mL/s F_ENO50) among COPD Phenotypes. *: $p < 0.05$ and **: $p < 0.01$ (Kruskal–Wallis test). FE-E: frequent exacerbators with emphysema. FE-CB: frequent exacerbators with chronic bronchitis. NE-CB/E: non-frequent exacerbators with chronic bronchitis or emphysema. ACOS: asthma-COPD overlap syndrome.

population, and also could be a source of bias. Another weakness of the study is the cross-sectional design, which does not allow the temporal stability of the marker to be asserted, an important fact in the evaluation of any biomarker in COPD, as previously observed [32].

In conclusion, the determination of F_ENO50 in COPD patients may help to improve the categorization of patients, detect patients with ACOS, and allow more personalized treatment. F_ENO50 values above 19 ppb should be included as diagnostic criteria of ACOS.

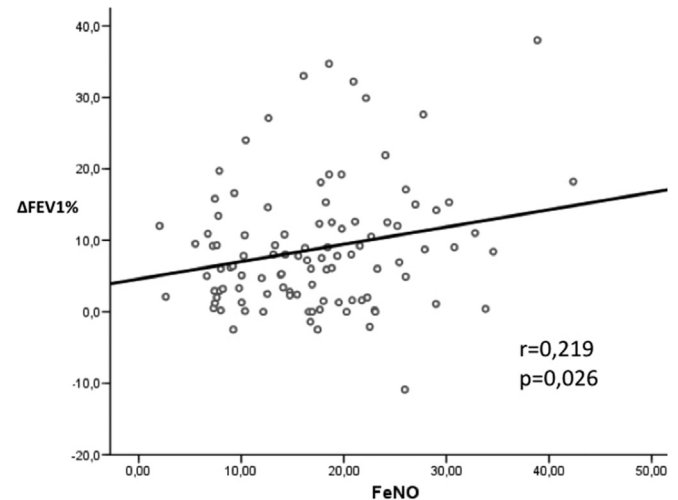


Fig. 5. Relationship between Bronchodilator Response to Salbutamol (expressed as ΔFEV₁% from baseline) and F_ENO50 Levels in COPD Patients.

Authors contribution

BAN had full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis, including and especially any adverse effects. BAN, PJR, ARS and ORR contributed substantially to the study design, data analysis and interpretation, and the writing of the manuscript.

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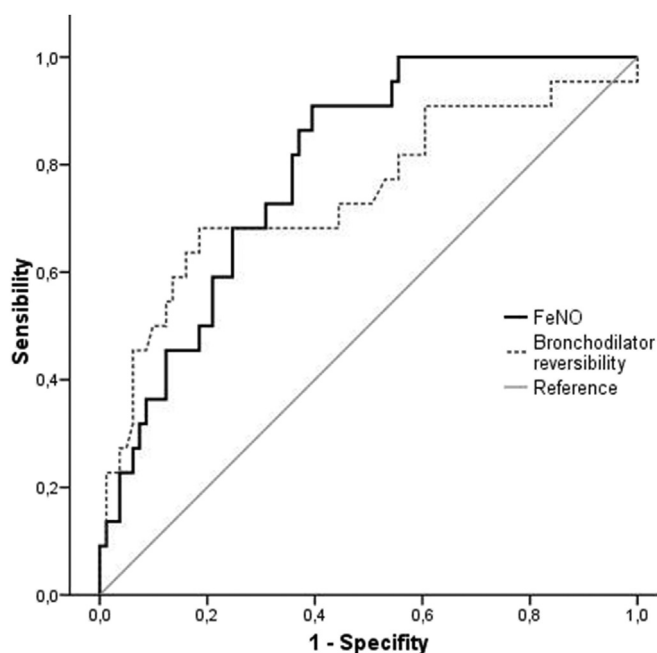


Fig. 6. –ROC (receiving operator characteristics) curves for diagnosing ACOS with FeNO_{50} levels and bronchodilator reversibility (Defined as improvement in $\text{FEV}_1 > 200$ mL and $> 12\%$ after 400 μg salbutamol). AUC FeNO_{50} : 0.793 (95% CI 0.70–0.88). AUC Bronchodilator reversibility: 0.746 (95% CI 0.61–0.87). Sensitivity (FeNO_{50} 19 ppb): 0.68. Specificity (FeNO_{50} 19 ppb): 0.75.

Conflict of interests

Dr. Alcázar reports personal fees from Novartis AG, personal fees from Boehringer–Ingelheim, personal fees from GSK, personal fees from Almirall, grants and personal fees from Menarini, personal fees from Astra–Zeneca, outside the submitted work.

Dr. Romero-Palacios has nothing to disclose.

Dr. Ruiz Sancho has nothing to disclose.

Dr. Ruiz Rodriguez has nothing to disclose.

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